Prospect Desk

Lead Intelligence Tool | Sales & Operations track | System documentation | Author: Anuj Patel

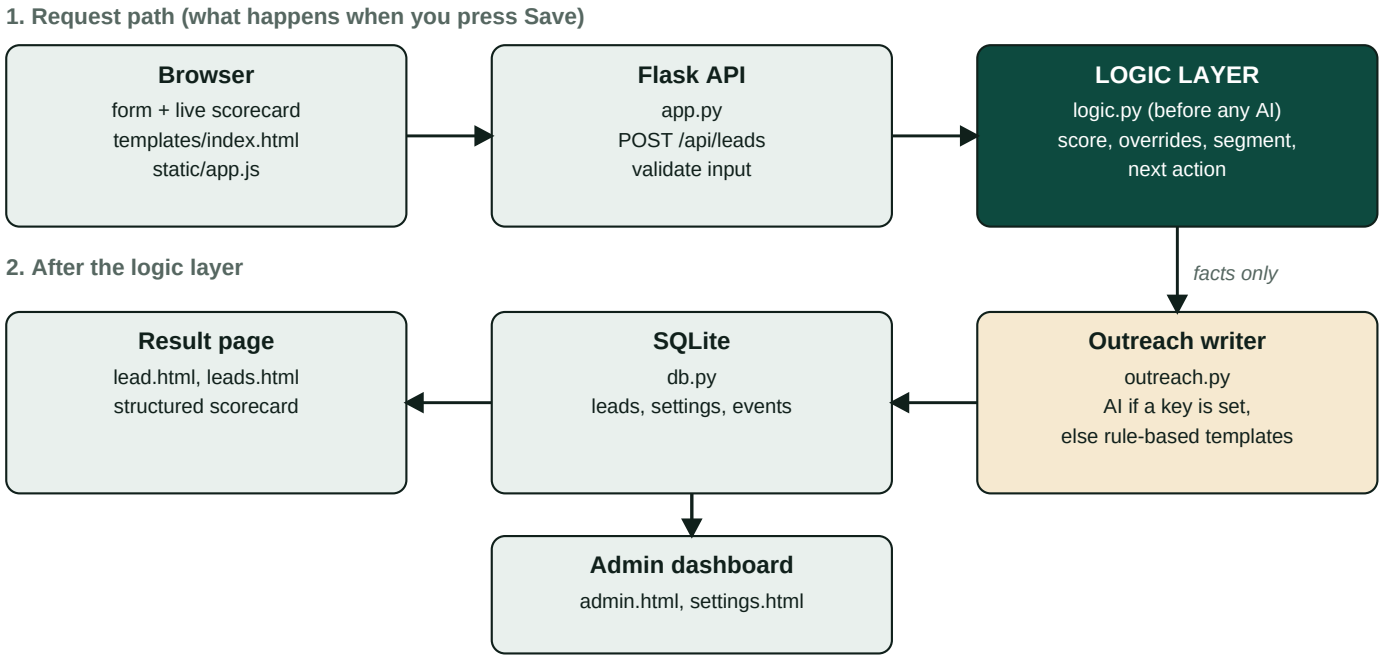
1. Overview

Prospect Desk helps a sales team decide which leads deserve a call. A rep enters a lead's role, company and what is known about them. The system scores the lead out of 100, explains every point, places it in a segment (Hot, Warm, Nurture or Park), names the next action with a reply time, and drafts a personalised outreach message. Every lead is stored in a pipeline and an admin dashboard shows how the system is being used.

Input	Lead name, role, company, industry, company size, source, days since last contact, free-text context
Output	Lead score (0-100), segment, next action with reply time, outreach message, plain-language reasons
Logic layer	Scoring rules, override rules, segmentation, action mapping (all deterministic, all run before any AI)
AI	Optional. Writes the outreach message only. A rule-based writer is used when no key is set or the AI fails a check.

Design principle. The decisions that matter (is this lead worth a call, and what should we do) are made by rules a sales manager can read, question and change. The AI is used for what it is good at, phrasing a message, and it never changes a score, segment or action.

2. System architecture



The browser sends JSON to Flask. The server validates it, runs the logic layer, asks the outreach writer for a message, stores the lead and returns its id. The browser then opens the saved lead as a structured page. The same logic layer also answers the live preview while the user types, so what the user sees while typing is exactly what will be saved.

Layer	File	Responsibility
Frontend	templates/, static/	Structured form, live scorecard, result page, pipeline, admin pages. No raw JSON is ever shown.
API and pipeline	app.py	Routes, validation errors as JSON, CSRF checks, admin sign-in, CSV export, calling each stage in order.

Layer	File	Responsibility
Logic layer	logic.py	Validation, scoring, overrides, segmentation, next-action mapping. Pure functions with no I/O, so they are easy to test.
Outreach writer	outreach.py	Template writer, optional LLM writer, output checks, fallback to templates.
Database	db.py	SQLite storage of leads, settings and an audit trail, plus the dashboard queries.

Key decisions

- **Logic before AI.** Scoring never depends on a model, so results are repeatable and explainable.
- **AI is optional.** With no API key the whole product still works. This also keeps cost at zero for a prototype.
- **One scorecard template.** The live preview and the saved page render the same partial (`_scorecard.html`), so they cannot drift apart.
- **Settings in the database.** The ideal customer profile and thresholds are edited from the admin page and stored, not hard-coded.
- **Audit trail.** Every score, AI result, status change, deletion and settings change is written to an events table.

3. Frontend implementation

The frontend is server-rendered Flask templates with one small script (`static/app.js`) for the live scorecard, form submission and CSV import. There is no front-end framework. Styles are in one CSS file with a light and a dark theme and a layout that works down to phone width.

Page	Route	Purpose
Score a lead	/	Structured form on the left, live scorecard on the right. Buttons: Save lead and write outreach, Fill an example, CSV import.
Lead	/lead/<id>	Saved scorecard, outreach message with copy button, status control, the facts and context that were entered.
Pipeline	/leads	All leads sorted by score, filter by segment, status and search.
Admin	/admin	Usage dashboard, latest leads, activity log (password protected).
Scoring settings	/admin/settings	Ideal customer profile, thresholds, sender details (password protected).

How the scorecard reads. The large number is the score. The coloured band shows the four segments with their thresholds, and the black marker shows where this lead falls. Under it, five rows show points earned against points possible, each with a one-line reason. A yellow note appears whenever an override rule changed the result. The box at the bottom is the next action.

Input handling. Required fields are checked on the server and the errors are listed together above the form, in plain language, with what the user typed kept in place. Context is limited to 1,500 characters with a live counter. The page tells users to enter made-up details, because the demo workspace is not private.

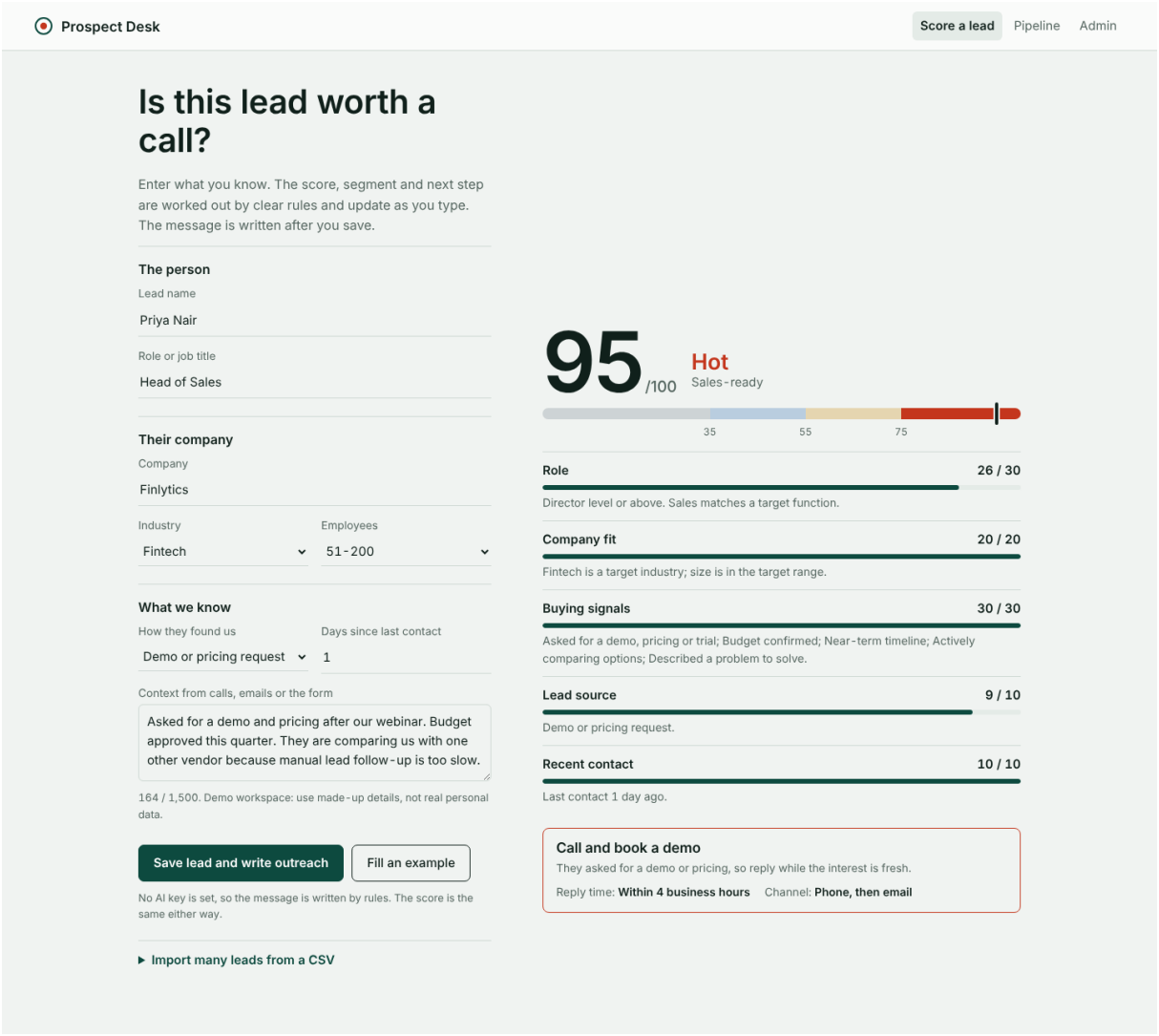


Figure 1. The score page. The scorecard on the right updates as the form on the left is filled in.

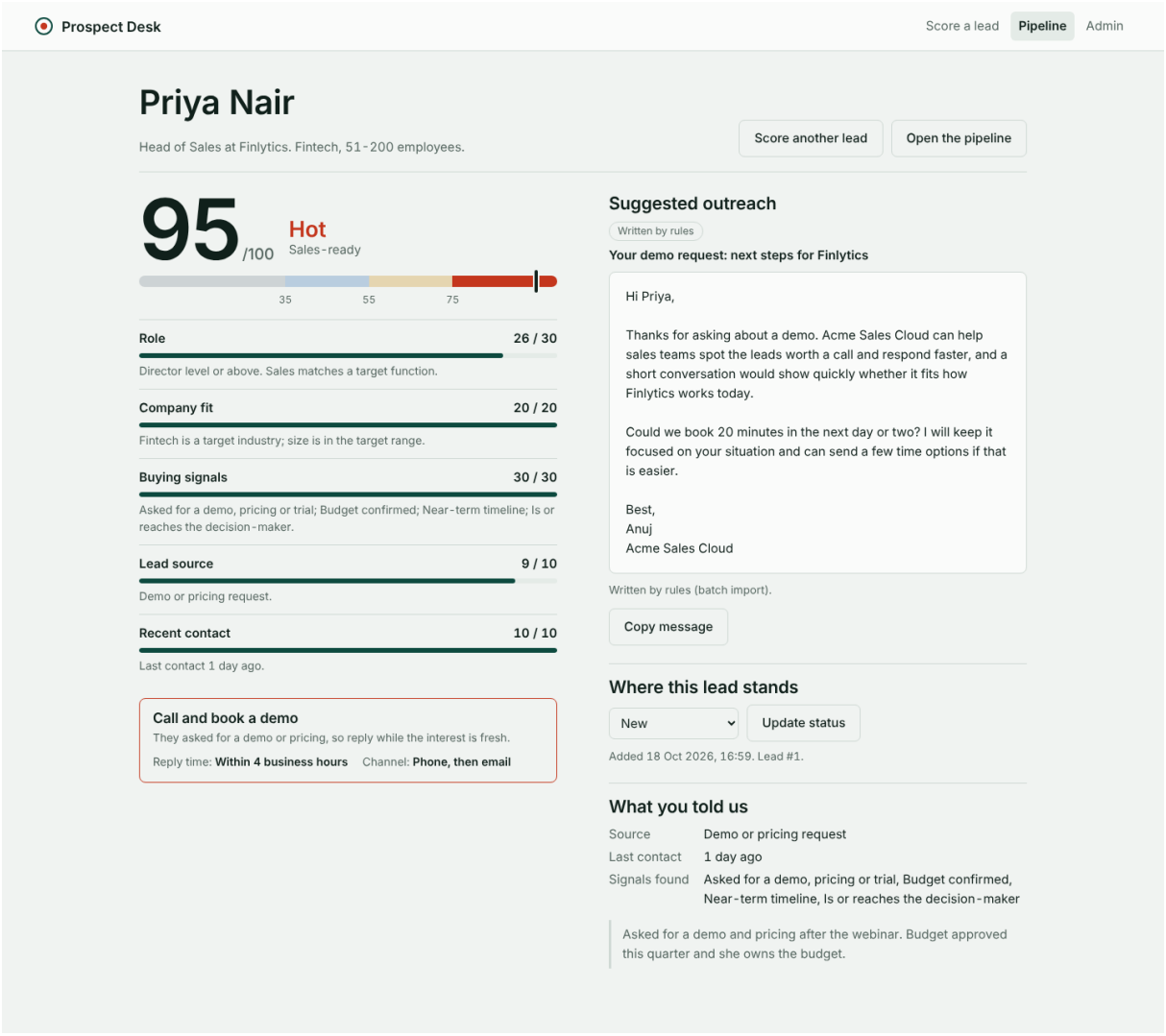


Figure 2. A saved lead: full scorecard, outreach message, status control and the original context.

4. Backend and logic layer

API

Endpoint	What it does
POST /api/preview	Validates loosely, runs the logic layer only, returns the rendered scorecard. No AI call, nothing stored.
POST /api/leads	Full pipeline: validate, score, write outreach, store, return the new lead id. Returns a list of errors with status 400 on bad input.
POST /api/batch	Imports up to 50 CSV rows. Each valid row is scored and stored with a rule-based message. Invalid rows are reported and skipped.
POST /lead/<id>/status	Changes a lead's status. Protected by a CSRF token.
GET /admin, /admin/settings, /admin/export.csv	Dashboard, settings and CSV export. Require the admin password.

Scoring model (100 points)

Category	Max	How it is calculated
Role	30	Seniority up to 20: executive or owner 20, director level 16, manager 10, individual contributor 5, unclear 4, student 1. Function fit up to 10: 10 if the title matches a target function, 3 if another known function, 2 if unreadable.

Category	Max	How it is calculated
Company fit	20	Industry 10 if it is a target industry, otherwise 3. Size 10 if in a target band, 6 if one band away, 2 if further.
Buying signals	30	Points for signals found in the context text (table below), capped at 30.
Lead source	10	Referral 10, demo or pricing request 9, event 7, webinar or content 6, website sign-up 5, outbound 4, purchased list 2.
Recent contact	10	0 to 3 days 10, 4 to 7 days 8, 8 to 14 days 6, 15 to 30 days 3, over 30 days 0. If no contact is logged yet, 5 (neutral).

Signals read from the context text

Signal	Points	Examples of what triggers it
Asked for demo, pricing or trial	+10	demo, pricing, quote, proposal, trial, pilot
Budget confirmed	+8	budget approved, have a budget, funding approved
Near-term timeline	+7	this quarter, next month, asap, urgent, go-live
Comparing options	+6	evaluating, comparing, shortlist, RFP, switching from
Described a problem	+5	struggling, manual, losing, churn, bottleneck
Is or reaches the decision-maker	+5	final say, sign-off, owns the budget, reports to the CRO
Came via a referral	+4	referred, introduced by, recommended by
Asked not to be contacted	-20	not interested, unsubscribe, stop emailing
Likely not a buyer	-12	just browsing, for a class project, competitor
Said there is no budget	-10	no budget, budget freeze
Low urgency	-6	maybe next year, not a priority

Penalties are limited to a total of -25. The final score is the sum of the five categories plus penalties, kept between 0 and 100.

Override rules. After scoring, four rules protect against misleading results. Each one that fires is shown on the lead page.

- **Asked not to be contacted:** the score is capped at 20 and the next action becomes Do not contact.
- **Student or intern:** the score is capped at 34 (Park), because the role is rarely the buyer.
- **Hot needs a buying signal:** a score of 75 or more is held at 74 if buying signals earned less than 10 points. A good profile alone is not enough.
- **Gone quiet:** a Hot lead with no contact for over 45 days is held at 74.

Segments and next actions. Defaults are Hot from 75, Warm from 55, Nurture from 35 and Park below 35. All three thresholds are editable in the admin settings.

Segment	Condition	Next action	Reply time
Hot	Asked for a demo or pricing	Call and book a demo	Within 4 business hours
Hot	Otherwise	Call, then send a tailored email	Within 1 business day
Warm	Contact is not senior or a decision-maker, no decision-maker signal	Ask for an introduction to the decision-maker	This week
Warm	Otherwise	Send a discovery email and propose a 20-minute call	Within 3 business days
Nurture	Any	Add to the nurture sequence	Review in 30 days

Segment	Condition	Next action	Reply time
Park	Any	No rep time for now	Re-score if status changes
Any	Asked not to be contacted	Do not contact	Today

Worked example. Elena Rossi, Chief Revenue Officer at a fintech of 51-200 people, came through a referral. Her context says a partner introduced her and that she mentioned slow follow-up. Role 30, company fit 20, buying signals 9, source 10 and recent contact 10 add up to 79, which would be Hot. Buying signals are under 10 points, so the override holds her at 74 (Warm) and the next action asks for a discovery call rather than a hot-lead call. This is the case that stops a good job title from inflating a lead.

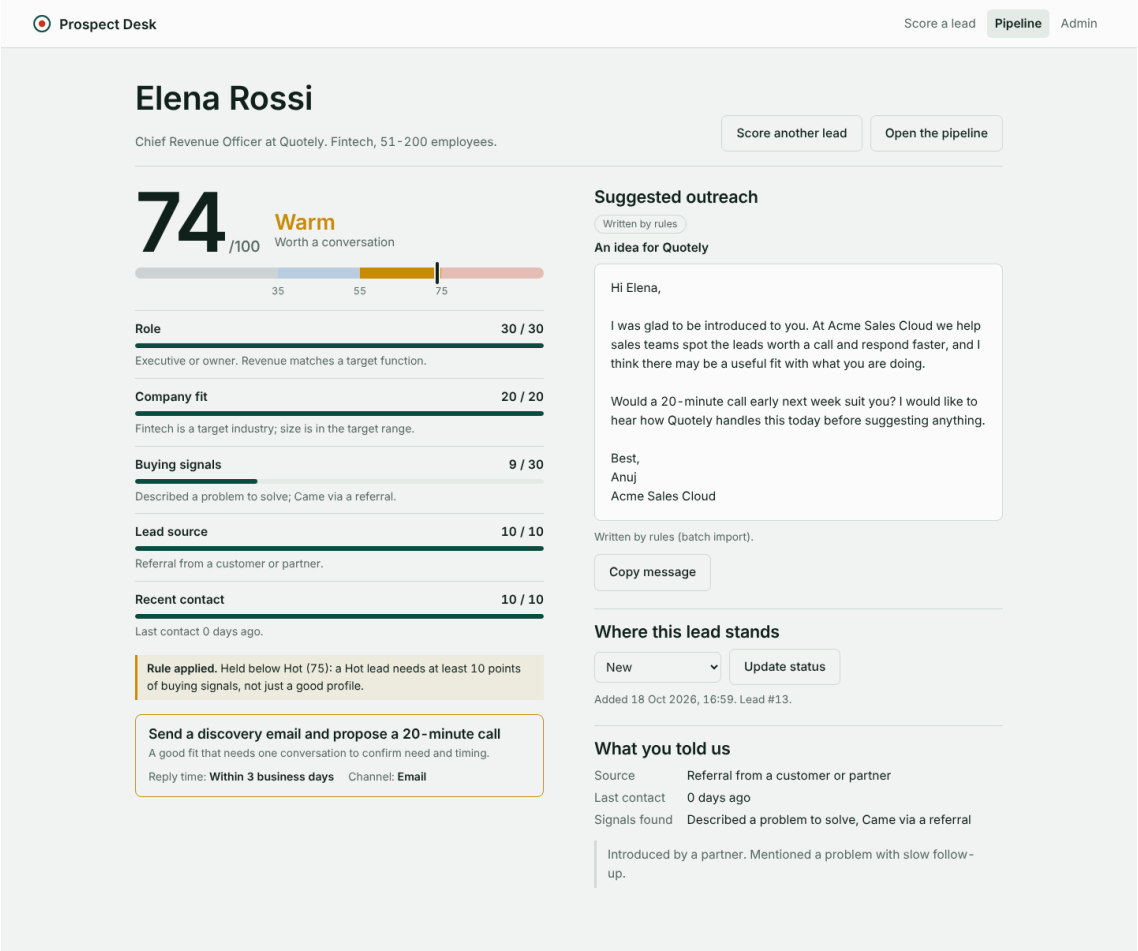


Figure 3. The override rule on a saved lead: 79 points of evidence held at 74, with the rule shown to the user.

Outreach writer and checks

`outreach.generate()` returns a message for Hot, Warm and Nurture leads and none for Park or do-not-contact leads. If an API key for Gemini, OpenAI or Anthropic is set, the model receives only structured facts about the lead, the segment and the next action, and the lead's free text is marked as untrusted data. The reply is then checked before it is accepted:

- length between 40 and 200 words, and a non-empty subject;
- no links, no unfilled placeholders, no unsupported claims such as "as we discussed" or "guaranteed";
- every number must appear in the lead data (a meeting length such as 20 minutes is allowed);
- must mention the lead or the company.

If the call fails, times out, returns bad JSON or fails any check, the rule-based template is used and the page says why. With no key at all, templates are used and labelled *Written by rules*.

5. Database design

SQLite with three tables. The scorecard (breakdown, signals, flags, overrides and thresholds) is stored as JSON on the lead, so an old lead always shows the score and reasons it had when it was scored, even if the settings change later.

Table	Key columns	Purpose
leads	id, created_at, name, role, company, industry, size, source, context, last_contact_days, score, base_score, segment, seniority, action_*, breakdown_json, signals_json, flags_json, overrides_json, thresholds_json, outreach_mode, outreach_subject, outreach_body, provider, model, tokens_in, tokens_out, latency_ms, cost_usd, status	One row per scored lead: inputs, full scorecard, outreach, AI usage and pipeline status.
settings	key (primary key), value (JSON)	Ideal customer profile, thresholds, sender name, company and offer. Defaults come from code and are overridden by stored values.
events	id, ts, type, lead_id, detail	Audit trail: lead_scored, outreach_ai, ai_failed, ai_rejected, status_changed, lead_deleted, settings_updated, admin_login.

Indexes are on `leads(segment)` and `leads(status)`, the two filters used by the pipeline page. Lead status moves through new, contacted, meeting booked, won or lost. The outreach mode is ai, template or none. All queries are parameterised. Deleting a lead removes its row and writes an event, and events are kept.

6. Admin dashboard

The admin area is behind a password (the ADMIN_PASSWORD setting). It answers three questions: how is the pipeline distributed, how is the AI behaving, and is the scoring tuned well.

Widget	What it shows	Why it matters
Key figures	Leads scored, average score, hot leads and share, AI vs rules drafts, AI fallbacks, AI cost and tokens	Quick health check
Leads by segment	Count per segment	If nearly everything is Hot or Park, thresholds need tuning
Pipeline status	Count per status	Shows how far leads move after scoring
Where leads come from	Count per source	Shows which sources produce leads
Rules that changed a result	How often each override fired	Shows how much the safeguards are doing
Outreach writer	Whether AI is on, provider and model, average AI time	Shows the current mode
Leads per day	Bar per day with data	Usage over time
Latest leads	Table with delete	Manage records
Activity log	Newest 25 events	Audit trail
Scoring settings	Edit target industries, sizes, functions, thresholds and sender	Tune the model without code changes
Export CSV	Download all leads	Use the data elsewhere

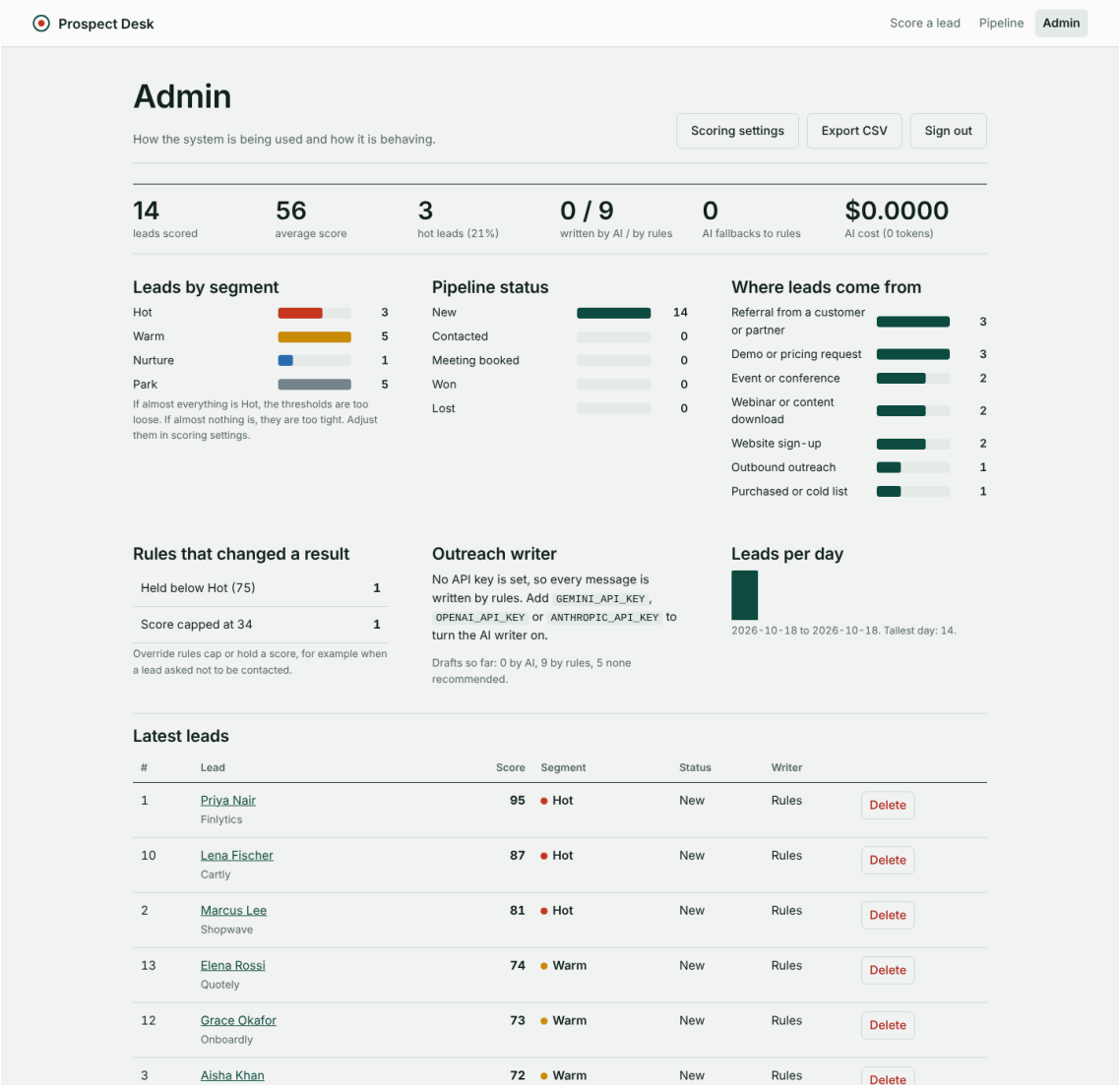


Figure 4. The admin dashboard (top of the page).

Security measures. Form posts carry a CSRF token. Admin password comparison is constant-time. Templates escape all user text. CSV exports prefix cells that could be read as spreadsheet formulas. Input lengths are limited, request size is limited, and the page warns while the default admin password is in use.

7. Git commit progression

Development was staged so that each stage has its own commit, in the order the system is built up.

#	Commit	Stage	What it added
1	`8f5dcd6`	Setup	README, requirements, config template
2	`ef062e4`	Frontend	Base layout, styles, input form, scorecard partial, live preview script
3	`1c3aa14`	Logic layer	Validation, 5-category scoring, override rules, segments, action mapping, sample leads
4	`6936e12`	AI / outreach	Template writer, optional LLM writer (3 providers), output checks, fallback
5	`354fc72`	Database	leads, settings and events tables, dashboard queries
6	`7df498d`	Backend integration	Routes, API, CSV import, lead and pipeline pages
7	`86ffc74`	Admin dashboard	Admin dashboard, scoring settings, sign-in
8	`this commit`	Tests and docs	14 tests, this PDF, screenshots

8. Testing, limitations and next steps

Tests. `tests/test_app.py` holds 14 tests. They cover hot-lead scoring and action mapping, the do-not-contact cap, the student cap, both Hot hold rules, validation messages, deterministic scoring, rejection of invented figures and links in AI text, fallback when the AI fails or is rejected, acceptance of a clean AI draft, the full request flow, admin sign-in with CSRF, the settings rules and CSV import. The pages were also checked in a real browser: validation errors, saving, the live scorecard, status change, CSV import, filters and a phone-width layout without sideways scrolling.

Known limitations

- The AI writer was tested with simulated model responses. It was not run against a live model during development because no API key was used, so the first run with a real key should be checked by hand.
- Buying signals are matched by keywords in English, so unusual phrasing can be missed. The reasons shown on every score make a miss easy to spot.
- The scoring weights are reasoned defaults, not fitted to real win and loss data. Recording outcomes (status won or lost) is the first step towards tuning them.
- SQLite on a host that resets its disk will lose leads. The pipeline and leads are visible to anyone with the link in this demo, so only made-up data should be entered.
- Single admin password, no per-user accounts.

Next steps. Compare score against won and lost status to adjust weights, add CRM import, add per-user ownership of leads, and add reminders when a reply time is missed.